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Supplemental Material

Impact of Reductions in Emissions from Major Source Sectors on Fine Particulate Matter–Related Cardiovascular Mortality

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Table S1. Correlation matrix between county-wise annual change in PM_{2.5} components.

Values on diagonal (shaded boxes) are the correlations between the baseline concentrations in 1990 and the annual change in concentrations; negative values indicate that counties with higher baseline concentrations in 1990 tend to have stronger decreases. Values below diagonal are cross-correlations between county-wise annual trends in the components, and values above diagonal are cross-correlations between the counties' baseline component concentrations in 1990. Larger values in cross-correlations indicate that concentrations of the components tend to correlate, either in the baseline concentration or annual reduction trend. Baseline and annual trend in total PM_{2.5} concentration is most strongly correlated to sulfates concentration. Elemental and organic carbon concentrations are the most highly correlated components.

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Table S4. Orthogonal two-factor analysis of mortality trends by PM_{2.5} component concentration. Mortality trends were measured in cardiovascular deaths (per 100,000 person-years) per Year; mortality risks were measured in cardiovascular deaths (per 100,000 person-years) per $\mu\text{g}/\text{m}^3$. We reported standard errors in parentheses and 95% confidence intervals in brackets. Regressions were weighted by inverse standard errors of baseline CV mortality rates and adjusted for covariates median household income, county-level percent non-white, age-standardized COPD mortality rates, baseline year 1990 CV mortality rates, and baseline year 1990 PM_{2.5} concentrations.

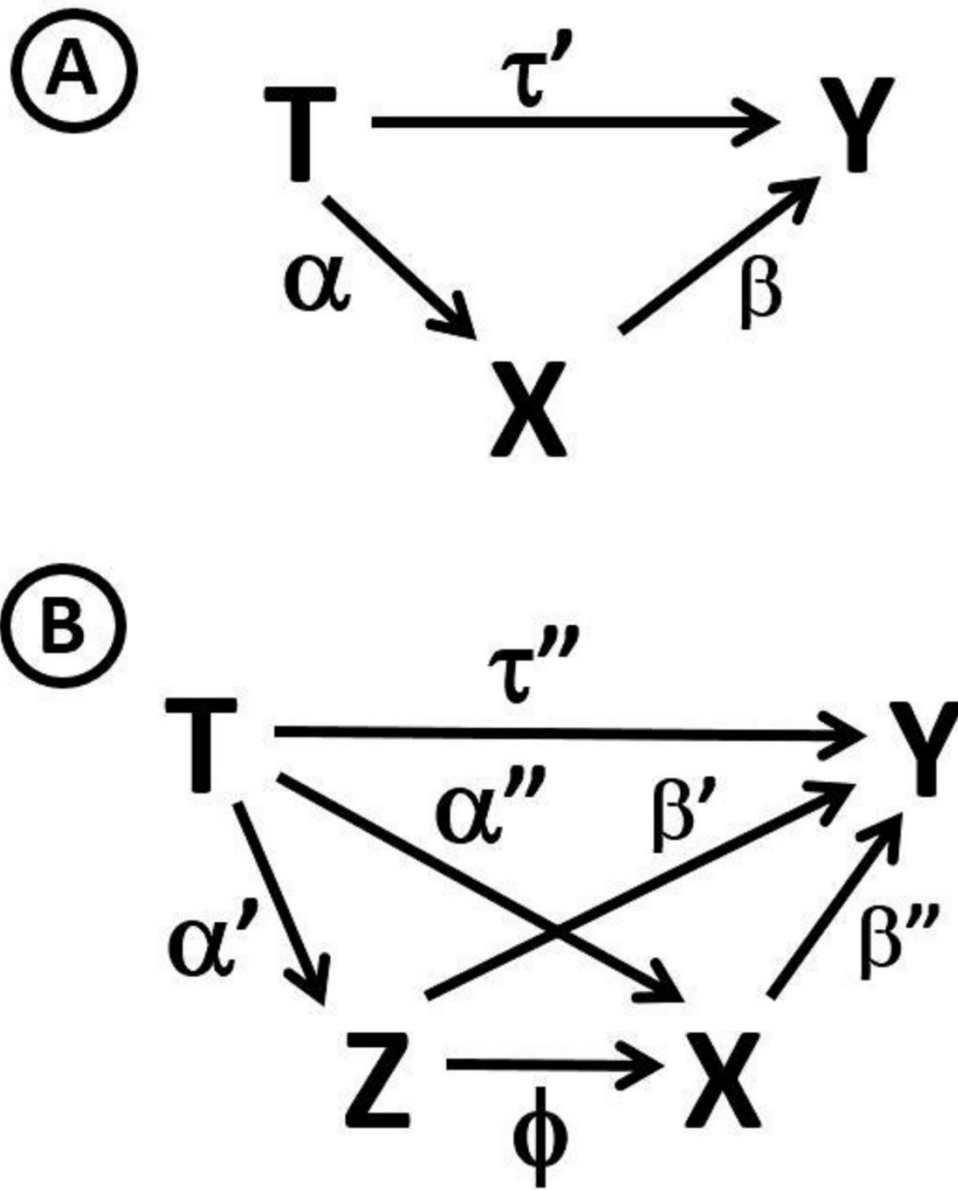


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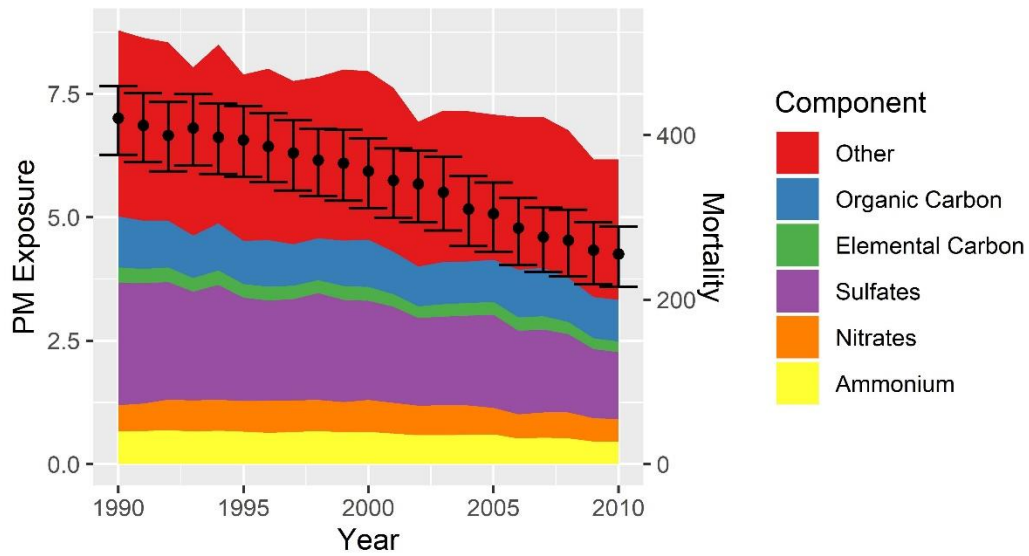
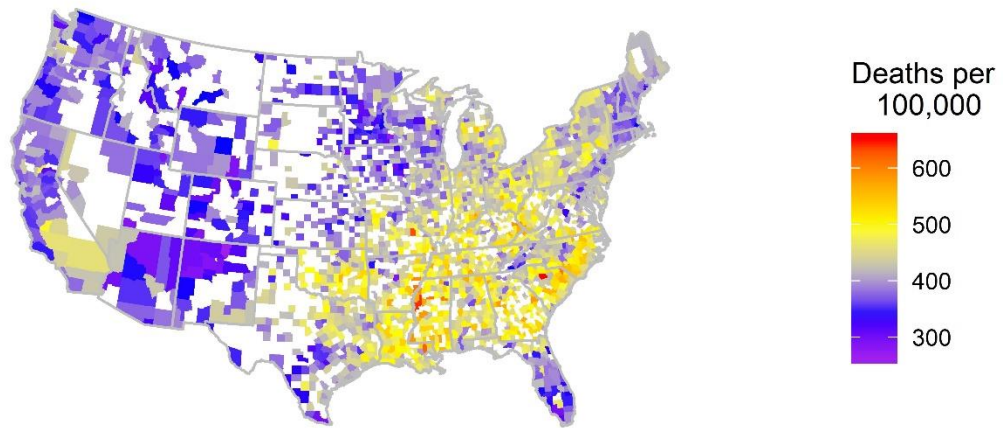


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Cardiovascular Mortality Rates in 1990



Annual Trend in Cardiovascular Mortality Rates

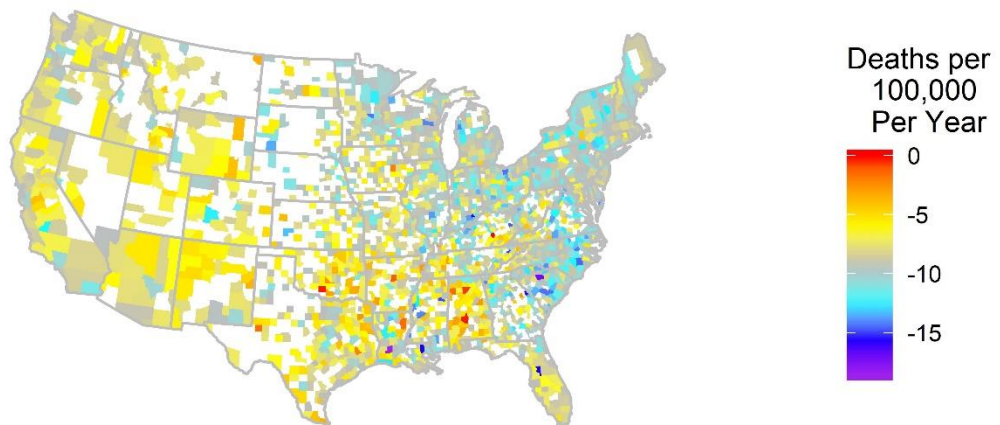
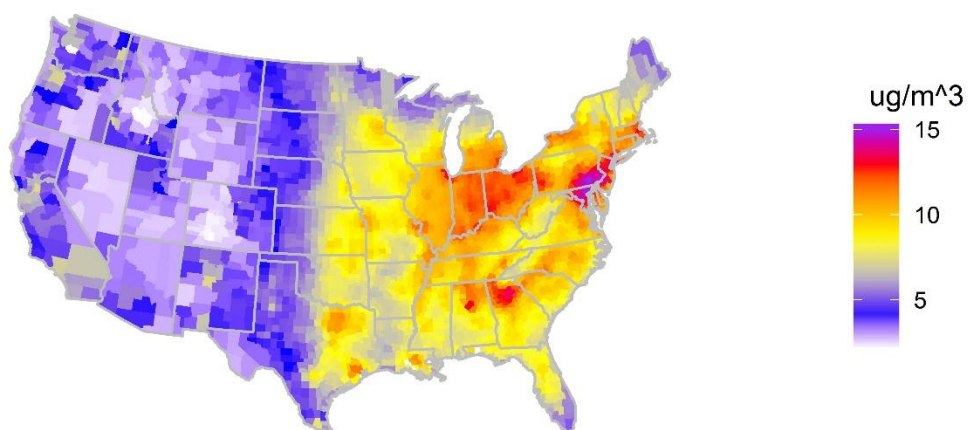


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Total PM 2.5 Concentrations in 1990



Annual Trend in Total PM2.5 Concentrations

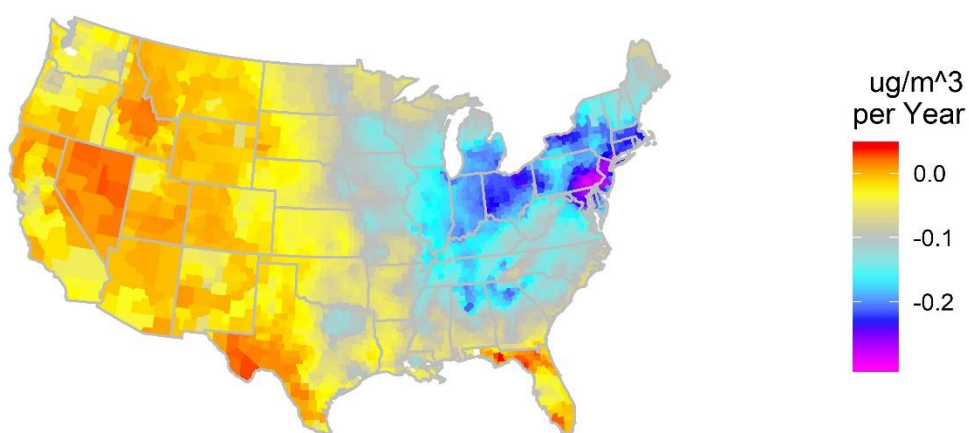


Fig. S4. Modeled annual average of daily concentration of total PM_{2.5} in $\mu\text{g}/\text{m}^3$ for all contiguous U.S. counties (top) in 1990 and (bottom) average annual change from 1990 to 2010.

	Nitrates	Sulfates	Elemental Carbon	Organic Carbon	Total PM _{2.5}
Nitrates	-0.855	0.526	0.562	0.426	0.760
Sulfates	0.485	-0.972	0.721	0.761	0.903
Elemental Carbon	0.513	0.703	-0.814	0.891	0.847
Organic Carbon	0.522	0.609	0.922	-0.495	0.842
Total PM _{2.5}	0.703	0.894	0.873	0.839	-0.914

Table S1. Correlation matrix between county-wise annual change in PM_{2.5} components. Values on diagonal (shaded boxes) are the correlations between the baseline concentrations in 1990 and the annual change in concentrations; negative values indicate that counties with higher baseline concentrations in 1990 tend to have stronger decreases. Values below diagonal are cross-correlations between county-wise annual trends in the components, and values above diagonal are cross-correlations between the counties' baseline component concentrations in 1990. Larger values in cross-correlations indicate that concentrations of the components tend to correlate, either in the baseline concentration or annual reduction trend. Baseline and annual trend in total PM_{2.5} concentration is most strongly correlated to sulfates concentration. Elemental and organic carbon concentrations are the most highly correlated components.

	Baseline Estimate in 1990	Mean Annual Change	Total Change between 1990 and 2010	Percent Change
Cardiovascular MR (deaths per 100,000 person-years)	416.9 (0.5) [389.9, 470.8]	-9.52 (0.04) [-9.95, -7.19]	-190.3 (0.8) [-199.1, -143.9]	-45.66% (0.02) [-16.31, -34.16]
Total PM_{2.5} ($\mu\text{g}/\text{m}^3$)	9.458 (0.023) [6.297, 10.085]	-0.1395 (0.0017) [-0.1526, -0.0478]	-2.790 (0.035) [-3.052, -0.956]	-29.5% (0.2) [-31.1, -15.3]
Nitrates ($\mu\text{g}/\text{m}^3$)	0.686 (0.003) [0.360, 0.816]	-0.0077 (0.0002) [-0.0082, -0.0021]	-0.155 (0.005) [-0.164, -0.043]	-22.6% (1.4) [-22.3, -10.4]
Sulfates ($\mu\text{g}/\text{m}^3$)	2.497 (0.007) [1.416, 3.160]	-0.0526 (0.0006) [-0.0707, -0.0168]	-1.052 (0.011) [-1.413, -0.336]	-42.1% (0.1) [-46.0, -23.9]
Elemental Carbon ($\mu\text{g}/\text{m}^3$)	0.423 (0.002) [0.202, 0.330]	-0.0059 (0.0001) [-0.0043, -0.0006]	-0.119 (0.002) [-0.085, -0.012]	-28.1% (0.5) [-26.7, -6.0]
Organic Carbon ($\mu\text{g}/\text{m}^3$)	1.121 (0.003) [0.646, 1.045]	-0.0089 (0.0002) [-0.0069, +0.0028]	-0.178 (0.005) [-13.8, +0.057]	-15.9% (1.1) [-14.5, +8.2]

Table S2: Unadjusted annual trends in cardiovascular mortality rate (deaths per 100,000 people) and PM_{2.5} concentrations: 2132 U.S. counties, 1990-2010. Mean estimates across all counties were reported, with standard errors in parentheses and interquartile ranges of county-wise estimates in brackets. The total change equals the annual change times twenty years. The percent change is the total change divided by the baseline 1990 estimate. Regressions in this table did not adjust for covariates.

	Total PM _{2.5}	Ammonium	Nitrate	Sulfate	Elemental Carbon	Organic Carbon	Other
Annual PM Trend (α)	0.134 (0.001) [0.132, 0.136]	0.0110 (0.0001) [0.0108, 0.0112]	0.0068 (0.0001) [0.0066, 0.0070]	0.0501 (0.0002) [0.0497, 0.0505]	0.00613 (0.00003) [0.00607, 0.00619]	0.0091 (0.0001) [0.0089, 0.0093]	0.0511 (0.0002) [0.0507, 0.0515]
Non-PM- Related CV Mortality Trend (τ')	8.674 (0.028) [8.618, 8.730]	8.561 (0.024) [8.513, 8.609]	8.810 (0.020) [8.770, 8.85]	8.667 (0.028) [8.611, 8.723]	8.828 (0.023) [8.782, 8.874]	9.203 (0.019) [9.165, 9.241]	8.927 (0.027) [8.873, 8.981]
CV Mortality Risk (β)	3.884 (0.161) [3.562, 4.206]	57.55 (1.53) [54.49, 60.61]	56.10 (1.39) [53.32, 58.88]	10.56 (0.44) [9.68, 11.44]	59.95 (2.50) [54.95, 64.95]	-0.805 (0.811) [-2.427, 0.817]	5.267 (0.390) [4.487, 6.047]
PM-Related CV Mortality Trend ($\alpha\beta$)	0.521 (0.022) [0.477, 0.565]	0.633 (0.017) [0.599, 0.667]	0.384 (0.011) [0.362, 0.406]	0.530 (0.022) [0.486, 0.574]	0.367 (0.015) [0.337, 0.397]	-0.007 (0.007) [-0.021, 0.007]	0.269 (0.020) [0.229, 0.309]

Table S3: Single-factor analysis of mortality trends by differences in PM_{2.5}. Mortality trends were measured in cardiovascular deaths (per 100,000 person-years) per Year; annual PM trends were measured in $\mu\text{g}/\text{m}^3$ per year; mortality risks were measured in cardiovascular deaths (per 100,000 person-years) per $\mu\text{g}/\text{m}^3$. We reported standard errors in parentheses and 95% confidence intervals in brackets. Regressions were weighted by inverse standard errors of baseline CV mortality rates and adjusted for covariates median household income, county-level percent non-white, age-standardized COPD mortality rates, baseline year 1990 CV mortality rates, and baseline year 1990 PM_{2.5} concentrations.

	Ammonium	Nitrate	Sulfate	Elemental Carbon	Organic Carbon	Other
Association between Component and Total PM _{2.5} (ϕ)	9.765 (0.024) [9.717, 9.813]	7.647 (0.046) [7.555, 7.739]	2.504 (0.005) [2.494, 2.514]	16.693 (0.039) [16.615, 16.771]	5.313 (0.024) [5.265, 5.361]	2.463 (0.003) [2.457, 2.469]
Non-PM-Related CV Mortality Trend (τ'')	8.696 (0.027) [8.642, 8.750]	8.803 (0.027) [8.749, 8.857]	8.614 (0.029) [8.556, 8.672]	8.753 (0.028) [8.697, 8.809]	8.280 (0.031) [8.218, 8.342]	8.726 (0.028) [8.670, 8.782]
Component Mortality Risk (β')	51.94 (1.61) [48.71, 55.16]	56.31 (1.49) [53.33, 59.29]	11.16 (0.44) [10.28, 12.04]	65.02 (2.69) [59.63, 70.40]	12.23 (0.87) [10.50, 13.97]	7.35 (0.39) [6.56, 8.13]
Non-Component Mortality Risk (β'')	-2.696 (0.253) [-3.202, -2.190]	0.067 (0.178) [-0.289, 0.423]	2.547 (0.284) [1.979, 3.115]	1.396 (0.278) [0.840, 1.952]	9.363 (0.244) [8.875, 9.851]	11.272 (0.404) [10.464, 12.080]
Component PM-Related CV Mortality Trend ($\alpha'\beta'$)	0.572 (0.018) [0.536, 0.608]	0.385 (0.011) [0.363, 0.407]	0.560 (0.022) [0.516, 0.604]	0.398 (0.017) [0.364, 0.432]	0.111 (0.008) [0.095, 0.127]	0.376 (0.020) [0.336, 0.416]
Non-Component, PM-Related CV Mortality Trend ($\alpha''\beta''$)	-0.071 (0.007) [-0.085, -0.057]	0.005 (0.014) [-0.023, 0.033]	0.022 (0.003) [0.016, 0.028]	0.044 (0.009) [0.026, 0.062]	0.793 (0.021) [0.751, 0.835]	0.093 (0.004) [0.085, 0.101]

Table S4. Orthogonal two-factor analysis of mortality trends by PM_{2.5} component concentration. Mortality trends were measured in cardiovascular deaths (per 100,000 person-years) per Year; mortality risks were measured in cardiovascular deaths (per 100,000 person-years) per $\mu\text{g}/\text{m}^3$. We reported standard errors in parentheses and 95% confidence intervals in brackets. Regressions were weighted by inverse standard errors of baseline CV mortality rates and adjusted for covariates median household income, county-level percent non-white, age-standardized COPD mortality rates, baseline year 1990 CV mortality rates, and baseline year 1990 PM_{2.5} concentrations.